Einstein's "happiest thought" showed how to do so. To understand his insight, imagine the year is 2050, you are the FBI's chief explosives expert, and you have just received a frantic call to investigate what appears to be a sophisticated bomb planted in the heart of Washington, D.C. After rushing to the scene and examining the device, your worst nightmare is confirmed: The bomb is nuclear and of such powerful design that even if it were buried deeply in the earth's crust or submerged in an ocean's depth, the damage from its blast would be devastating. After gingerly studying the bomb's detonation mechanism you realize that there is no hope to disarm it and, furthermore, you see that it has a novel booby-trap feature. The bomb is mounted on a scale. Should the reading on the scale deviate from its present value by more than 50 percent, the bomb will detonate. According to the timing mechanism, you see that you have but one week and counting. The fate of millions of people rests on your shoulders—what do you do?

Well, having determined that there is no safe place anywhere on or in the earth to detonate the device, you appear to have only one option: You must launch the device into the depths of outer space where its explosion will cause no damage. You present this idea to a meeting of your team at the FBI and almost immediately your plan is dashed by a young assistant. "There is a serious problem with your plan," your assistant Isaac begins. "As the device gets farther from the earth, its weight will decrease, since its gravitational attraction with the earth will diminish. This means that the reading on the scale inside the device will decrease, causing detonation well before reaching the safety of deep space." Before you have time to fully contemplate this criticism, another young assistant pipes up: "In fact, come to think of it, there is even another problem," your assistant Albert says. "This problem is as important as Isaac's objection but somewhat more subtle, so bear with me as I explain it." Wanting a moment to think through Isaac's objection, you try to hush Albert, but as usual, once he begins there is no stopping him.

"In order to launch the device into outer space we will have to mount it on a rocket. As the rocket accelerates upward in order to penetrate outer space, the reading on the scale will increase, again causing the device to detonate prematurely. You see, the base of the